MANJU SRI S

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PORTFOLIO

About Me

Final-year Electronics and Communication Engineering student with a strong foundation in Embedded Systems, IoT, Artificial Intelligence, and Machine Learning. Experienced in microcontroller programming, sensor integration, and intelligent monitoring systems, with a keen interest in developing innovative engineering solutions and research-oriented technologies.

Education

B.E-ECE (2023-2027):
Sri Eshwar College of Engineering **CGPA 8.3**

HSC (2022-2023):
Barathi Vidya Bavan Higher Secondary School **92%**

SSLC (2020-2021):
Vighneshwar Vidhya Mandhir -SSLC **PASS**

Skills

- **Programming:** C, C++, Embedded C, Python, ALP
- **Microcontrollers:** Intel 8051, ESP32, STM32
- **Protocols:** UART, I2C, SPI, CAN, MQTT
- **Wireless Technology:** BLE, WiFi , Zigbee
- **Tools:** Keil, STM32CubeIDE, Arduino IDE, Matlab, VS Code, Github, Canva
- **Hardware Integration:** Sensor Integration, Actuators,ADC&DAC,PCB Design
- **Testing Instruments:** Multimeter,oscilloscope,Signal Generator
- **Operating System**: FreeRTOS, Basic Linux

Achievements

- Secured 2nd Position in intercollege event 'Circuit Digest' held as a part of of KRISHNECS 2K24 – A National Level Technical Symposium at Sri Krishna College if Engineering and Technology.

Research Publication:

- A Scalable and Intelligent Approach to Optimizing EV Charging Infrastructure-IEEE-ICEESI **2026**

Certifications

Mastering Data Structures & Algorithms using C and C++	Udemy| 2024
C Programming and Assembly Language	**NPTEL| 2025**
Full-Stack Development	**Simplilearn | 2025**
Introduction to Cyber Security	**Simplilearn | 2026**
SQL(Basic) – Standard	**Skillrack | 2024**

Internship

- **ARTIWIZ TECHNOLOGIES - EMBEDDED SYSTEMS & IOT (2025)**
  Developed and optimized real-time applications on Intel 8051 and STM32F407VGT6 using Bare Metal C, HAL, and FreeRTOS.

- **CENTRAL SCIENTIFIC INSTRUMENTS ORGANISATION (CSIR- CSIO) (2025)**
  Supported research and development activities in embedded systems and IoT by applying core electronics principles in real-time sensor integration and hardware deployment.

Projects

- **AI-Enabled Agro-Climate Intelligence & Adaptive Farm Decision Support System:**
  Developed an AI-powered agro-climate forecasting system using TFT to predict crop yield and environmental parameters. Built a battery-operated IoT system with sensor integration and real-time dashboard visualization.
  Tech Stack: *Machine Learning , Time-Series Forecasting , Temporal Fusion Transformer (TFT), Data Analysis , IoT Sensors*

- **Predictive Analytics and AI-Optimized Battery Management System for Electric Vehicles:**
  Engineered an IoT-enabled Battery Management System using ESP32 with real-time electrical and thermal monitoring. Integrated machine learning-based predictive analytics for battery health assessment and performance optimization.
  Tech Stack: *ESP32 , Current and Temperature sensors , ML (Random Forest) , SOC/SOH Estimation , Cloud (MQTT)*

Technical Focus

- Intelligent Embedded Systems
- Edge AI & TinyML
- Sustainable Energy Technologies